INCH-POUND

A-A-54966
15 July 1994

COMMERCIAL ITEM DESCRIPTION

WAX, BASEPLATE, DENTAL

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Federal Specification U-W-135B.

This Commercial Item Description covers dental baseplate wax suitable for forming occlusal rims and setting up artificial teeth

Salient characteristics:

Classification. The baseplate wax shall be furnished in the following types and classes, as specified:

Type I Medium wax for temperate weather.

Class 1 - Sheet form.
Class 2 - Ribbon form.

Type II. Hard wax for hot weather.

Class 1 - Sheet form.

Material. Type I, medium wax shall be composed essentially of high quality, synthetic waxes with a maximum of 7.0 percent beeswax, and shall be free from particulate fillers and liquid diluents.

Type I , hard wax shall consist essentially of mineral waxes (such as paraffin or microcrystalline), plant waxes, and beeswax with resins as an additive. A maximum of 1 percent of filler or opacifier is permissible.

Materials used in the formulation of the wax shall be of high quality and suitable in all respects for the purpose intended. The wax shall be free from extraneous materials such as dirt or lint, shall contain no harmful impurities, and shall have no odor or taste other than that characteristic in waxes normally used for this purpose.

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Description. The baseplate wax shall be suitable for forming occlusal rims and for setting up artificial teeth in denture construction. The wax shall be furnished clean and shall be suitable for use in the mouth in trial dentures. The wax shall have a smooth surface appearance. Type I wax shall be a medium wax suitable for use in temperate weather. Type II wax shall be a hard wax suitable for use in hot weather. Class 1 wax shall be furnished in sheet form and Class 2 wax shall be furnished in ribbon form. Wax shall be furnished in the types and classes as specified.

Shape and dimensions.

Class 1 wax. The wax shall be provided as individual sheets or scored multiple sheets. Individual sheets shall measure 5-7/8 ± 1/8 inches in length, 2-7/8 ± 1/16 inches in width, and 0.050 ± 0.005 inches in thickness. Multiple sheets shall measure 8-15/16 ± 1/8 inches in width, 5-7/8 ± 1/8 inches in length, and 0.050 ± 0.005 inches in thickness. Each multiple sheet shall have two scored lines so that when the sheet is broken, three sheets are obtained. Each of the separated sheets shall comply with the dimensions for an individual sheet.

Class 2 wax. The wax shall be furnished in ribbon form in a continuous roll measuring 5-1/2 ± 1/2 inches in diameter. The roll shall be wound around a sturdy roller and placed in a dispensing box with a side slot to permit ready withdrawal of the desired length of wax. The ribbon of wax shall measure 1-1/2 ± 1/16 inches in width and 0.050 ± 0.005 inches in thickness.

Weight. Type I, class 1 shall be a minimum of 5.0 pounds of wax. Type I, class 2 shall be a minimum of 1.0 pound of wax. Type II, class 1 shall be a minimum of 0.5 pound of wax.

Color. The color of the baseplate wax shall be pink to simulate appearance of gingival and alveolar tissues.

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Working properties. The wax shall have the necessary working properties for proper formation of occlusal rims and trial denture set ups. The wax shall be flexible and shall trim easily without chipping or flaking. When gently softened over the flame of a bunsen burner, the wax shall become moldable without crumbling, granulating, flaking, or laminating. When spot warmed with a heated wax spatula, the wax shall soften slightly, but not to a degree that the wax flows from the spatula. When spot heated with a fine flame point, the wax shall flow to produce a smooth glossy surface without loss of its molded form or anatomical detail. The wax shall cool quickly to a firmness which will render it capable of:

a being carved with a sharp knife without dragging or sticking.
b retaining porcelain and plastic teeth in proper position.

Physical properties.

Residue and coloring material. The molten wax shall show no color separation or insoluble substance when tested as specified below.

Flow. The wax shall conform to these flow limits when tested as specified below.

Flow limits.

Temperature ± 0.1°C	Percent flow (Type I)		Percent flow (Type II)	
	Minimum	Maximum	Minimum	Maximum
23.0	0.0	1.0	0.0	0.3
37.0	45.0	85.0	0.0	1.2
45.0	60.0	85.0	20.0	44.0

Linear shrinkage. The linear shrinkage of the wax shall not exceed 0.60 percent when tested as specified below.

Breaking point. At no point on the wax specimen shall there be a complete fracture after 10 seconds, when tested as specified below. A slight craze after 10 seconds will be permitted.

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Adhesion on storage. The adhesion on storage of the wax shall be such that when tested as described below, there shall be no evidence of damage to the wax surfaces that were in contact with each other or with the paper, where separating paper is used, and the wax and paper surfaces shall separate cleanly and readily.

Workmanship. The baseplate wax shall be free from defects which detract from its appearance or may impair its serviceability.

Unit. Box (BX).

For Type I, class 1, one box containing 5.0 pounds of wax, as specified, constitutes one unit.

For Type I, class 2, one box containing 1.0 pound of wax, as specified, constitutes one unit.

For Type II, class 1, one box containing 0.5 pound of wax, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one box of wax as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one box of wax. Sampling shall be inspection level S-2 with an AQL of 1.0.

Residue and coloring material test. An upper base plate of plastic shall be constructed using acceptable dental procedures. Plastic teeth shall be waxed on one side and porcelain teeth shall be waxed on the other side of the wax model denture. The wax model denture shall be constructed using the wax (Type I or II) being tested. The wax model denture shall be flasked in the usual manner using dental plaster or stone or both. The flask shall be allowed to set on the bench for at least two hours after the final set of the plaster. It shall then be placed into a water bath at $50^{\circ} \pm 5^{\circ}C$ for ten minutes, at which time the flask halves shall be separated and the bulk of the base plate shall be stripped out. The flask shall then be flushed with three applications of a boiling solution of synthetic household detergent (tablespoon to 1 pint of water) and rinsed with clean boiling water. The wax shall not leave a visually detectable residue on the porcelain or plastic teeth nor shall the coloring material separate from the wax or impregnate the plaster.

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Flow test.

Preparation of specimens. A sufficient number of sheets or rolls of wax shall be broken into pieces and placed in an iron crucible of 50 milliliter (ml) capacity. The crucible shall be placed 130 millimeters (mm) below a 250-watt infrared lamp bulb. The stirred wax shall be heated to a temperature of 75° ± 5°C until melted throughout and then shall be poured into a mold. The mold shall be a flat, stainless steel metal plate 6.0 mm thick, containing four holes 10.0 mm in diameter. The mold shall be preheated to a temperature of 55° ± 5°C and placed on a smooth glass slab (150 mm long, 75 mm wide, 20 mm thick) which shall have been preheated to the same temperature. After the mold has been overfilled with material, a smooth, flat, tinfoil-covered glass plate, preheated to 55° ± 5°C, shall be placed on top of the mold. A load of 90 newtons (N) shall be applied to the top of the tin-foil glass plate for 10 minutes. Then the mold shall be placed in water at a temperature of 10°C for 10 minutes. The upper tin-foiled plate shall then be removed and the excess material shall be trimmed away. This can be accomplished by using a metal scraper with the top surface of the mold as a guide. The mold may then be removed from the glass slab by gently tapping the side of the mold. The specimens shall be removed from the mold by chilling in water for 10 minutes at 10°C, and shall be stored at room temperature, 23° ± 2°C, for 24 hours before testing.

Method of test. The original length of the specimen shall be determined using a metric micrometer caliper. Four measurements shall be made around the circumference and one measurement shall be made on the center of the specimen. The measurements shall be recorded to the nearest 0.005 mm. The specimen and flow testing instrument shall be placed in a water bath and held at the testing temperature for 20 minutes prior to testing. The flow testing instrument consists of the following: (A) a weight; (B) a nonconducting shaft; and (C) a brass platen. The total weight in air of these three components is 19.6 N. The weight (A) shall be separated a minimum of 75 mm from the brass platen on the shaft. This shaft shall be of hard rubber or of a similar poor thermal conductor to avoid loss in heat due to conduction. The diameter of the brass platen which touches the specimen shall be no less than 50 mm. The thickness of the brass platen shall not exceed 6.5 mm. The temperature in the bath shall be controlled to within ± 0.1°C of the required temperature. A thin sheet of waterproof cellophane shall be placed between the instrument and each end of the specimen. The bottom of the specimen shall be 51 mm below the surface of the water in the bath. A constant axial load of 19.6 N shall then be applied to the specimen for 10 minutes, after which the specimen shall be removed and cooled in

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air to room temperature. The cellophane shall be stripped off, and the final length shall be determined. The flow, as evidenced by the change in length, shall be reported as percentage of the original length.

Linear shrinkage test. Specimens for determination of linear shrinkage shall be prepared in the same way as the flow test specimens. Linear shrinkage shall be determined on a specimen 200 to 300 mm in length and with equipment sufficiently accurate to insure correct results within plus or minus 0.05 percent. The wax shall be chilled from 40.0° ± 0.1°C to 25.0° ± 0.1°C and the linear shrinkage determined. The percent of linear shrinkage shall be the difference in specimen length at the two temperatures, divided by the length at 40°C, multiplied by 100.

Breaking point test. At a temperature of 21° ± 1°C for Type I wax and 27° ± 1°C for Type II wax, the baseplate wax shall be bent at a uniform rate for ten seconds until it is bent double, with the two halves of the specimen in contact throughout their entire length. The bend shall be perpendicular to the longer dimensions of the specimen.

Adhesion on storage test. Three specimens of wax and a specimen of separating paper, where such is used, having dimensions of 5 cm wide and 7.5 cm long shall be prepared. The specimen shall then be placed between two platens 5 cm wide, 6 cm long, and 0 5 cm thick in such a manner that 1.5 cm of each wax specimen projects from one end of the platens, and the separating paper, where used, is between two of the wax specimens. The assembled specimens shall then be placed in an oven maintained at a temperature of 40° ± 1.0°C on a flat rigid surface and a weight of 13.2 N, including the weight of the top plate, shall be applied. At the end of 24 hours the specimens shall be removed and allowed to cool to room temperature. At the end of two hours the specimens shall be separated by opening from the overlapping ends, and the surfaces in contact with other specimens shall be examined for evidence of damage.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

NOTE: The following National Stock Numbers and their respective Item Identifications are covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION	TYPE	CLASS
6520-00-449-5773	WAX, DENTAL, Baseplate, Medium, 1 pound	I	2
6520-00-567-0150	WAX, DENTAL, Baseplate, Hard, 1/2 pound	II	1
6520-00-842-1820	WAX, DENTAL, Baseplate, Medium, 5 pounds	I	1

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6520-2556
Location: paul/baseplat/dawne

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

   The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54966	940715

3. DOCUMENT TITLE

WAX, BASEPLATE, DENTAL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: DPSC-MSE	(1) Commercial (215)7372870	(2) AUTOVON 444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403, Falls Church VA 22041 3466 Telephone (703) 756 2340 AUTOVON 289 2340	

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